

OMAR REHAN

AI / ML Engineer · Data Analyst · Full-Stack Developer

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#) | [Hugging Face](#) | [Kaggle](#) | [Medium](#) | [Tableau](#)

PROFESSIONAL SUMMARY

AI/ML Engineer, Data Analyst, and Full-Stack Developer with a B.Sc. in Computer Science. Experienced building end-to-end systems — from data pipelines, model training, and BI dashboards (Power BI, Tableau) to production-ready full-stack web applications. Skilled in computer vision, deep learning, statistical analysis, and modern software engineering.

EDUCATION

Arab Academy for Science, Technology & Maritime Transport (AASTMT)

Oct 2019 – Feb 2024

B.Sc. in Computer Science · Graduation Project: AI-Powered Smart Classroom Automation — TensorFlow Lite CNN on Raspberry Pi; Flutter/Bluetooth IoT control of lighting, HVAC & projectors

PROJECTS & EXPERIENCE

SIVED: SAR Satellite Vehicle Detection · YOLOv8-ORB · PyTorch · OpenCV · Gradio · Hugging Face 2026

- 98.8% mAP@0.5 — fine-tuned YOLOv8m-ORB on 12,013 annotated SAR images with SAR-specific augmentation (disabled HSV, intensity variation), converging in 70 epochs on Tesla T4.
- Deployed Gradio app on Hugging Face Spaces with Gaussian heatmaps, ORB visualization, and frame-by-frame video detection.

Superstore Sales Analytics · Python · SQL (PostgreSQL) · Power BI · Scikit-learn · Streamlit 2026

- Analyzed 9,986 transactions (\$2.3M revenue); identified \$135K in net losses from excess discounting (>20%) and unprofitable Furniture categories (Tables/Bookcases).
- Engineered a PostgreSQL star schema (37 queries) for RFM segmentation & CLV; implemented Holt-Winters forecasting, K-Means clustering, and isolation forest anomaly detection; built a live Streamlit dashboard.

RehanPulse: Real-Time Developer Dashboard · Next.js 14 · TypeScript · Firebase · Zustand · Three.js 2026

- Full-stack hub: GitHub GraphQL API (HMAC-SHA256 webhooks), Vercel REST & Firestore via SSE; AES-256 GCM token encryption, Playwright E2E tests, Storybook component library.
- Pulse AI chatbot (Hugging Face Inference API, streaming SSE, live context injection); macOS frosted-glass UI with Three.js particle landing page and Framer Motion transitions.

Selected AI Projects · PyTorch · TensorFlow/Keras · Unsloth · LoRA/PEFT · TRL · Gradio 2025

- AstroVision:** Fine-tuned Meta Llama 3.2-11B Vision on 250 astronomy image-caption pairs (4-bit quantization, LoRA r=16, peak 10 GB GPU); BLEU & ROUGE-L evaluation; full walkthrough on Medium.
- Brain Tumor MRI:** InceptionV3 transfer learning with class-balanced augmentation, tf.data prefetching, Grad-CAM explainability, and multi-class AUC; deployed on Hugging Face Spaces.

IT Support & Data Management Specialist | Egyptian Army · Electronic Warfare Dept · North Sinai Apr 2024 – May 2025

- Managed digital records, database maintenance, and network infrastructure; contributed to software development, script automation, and AI integration initiatives.

TECHNICAL SKILLS

AI / ML & Vision: PyTorch, TensorFlow/Keras, YOLOv8-ORB, OpenCV, Hugging Face Transformers, Unsloth, LoRA/PEFT, LangChain, Grad-CAM, TFLite, Gradio, Scikit-learn

Data Analytics & BI: Power BI (DAX, Power Query), Tableau, Excel (Power Pivot, Pivot Tables), Orange Data Mining, PostgreSQL, MongoDB, MS SQL Server, Star Schema Design

Full-Stack Dev: Next.js 14, React, TypeScript, Node.js, FastAPI, Django, Flask, Firebase, Supabase, PostgreSQL, Docker, AWS, Vercel, REST, GraphQL, Tailwind CSS, Three.js, Framer Motion

Languages & Tools: Python, JavaScript/TypeScript, SQL, HTML/CSS, GitHub, Google Colab, Raspberry Pi, IoT, VS Code

CERTIFICATIONS

AI Associate & IoT Associate — Digital Hub (IBM-based) · Jan 2023 | CCNAv7: Enterprise Networking, Security & Automation · Switching, Routing & Wireless Essentials · Introduction to Networks (2022–2023) | Soft Skills — HRDC Cairo · Aug 2021

COMPETITIONS & LANGUAGES

Hult Prize — Regional Finals Competitor, Cairo Impact Summit (UN-affiliated) · Apr 2021 | **Arabic:** Native | **English:** Full Professional Proficiency